

Building Reliable AI Through Explainable AI-Enhanced Training Pipelines for Large Language Models

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Introduction

XAI or Explainable AI during training is motivated by the idea that explainability should not only diagnose a model after learning, but also “**shape what the model learns**”.

Motivation

- Improve model reliability
- Encourage “right-for-the-right-reasons” behavior
- Improve scientific validity
- Support domain-expert validation
- Improve alignment with task objectives
- Support regulatory and high-stakes deployment
- Reduce cost and energy by improving debugging and error analysis

Background

Several methods inject explanations, rationales, or interpretability constraints **during training or fine-tuning**. The strongest examples are in **instruction tuning, rationale supervision, process supervision, and explanation regularization**.

- RL for reasoning traces
- Constitutional AI / critique-and-revision training
- Explanation regularization
- Human-rationale supervision
- Rationale distillation

A compact training objective could be written as:

$$\mathcal{L}_{total} = \mathcal{L}_{task} + \lambda \mathcal{L}_{XAI} + \gamma \mathcal{L}_{science}$$

where:

$\mathcal{L}_{task}$ measures next-token prediction, classification, QA, or instruction-following performance.

$\mathcal{L}_{XAI}$ penalizes explanations that disagree with expert rationales or expected feature importance.

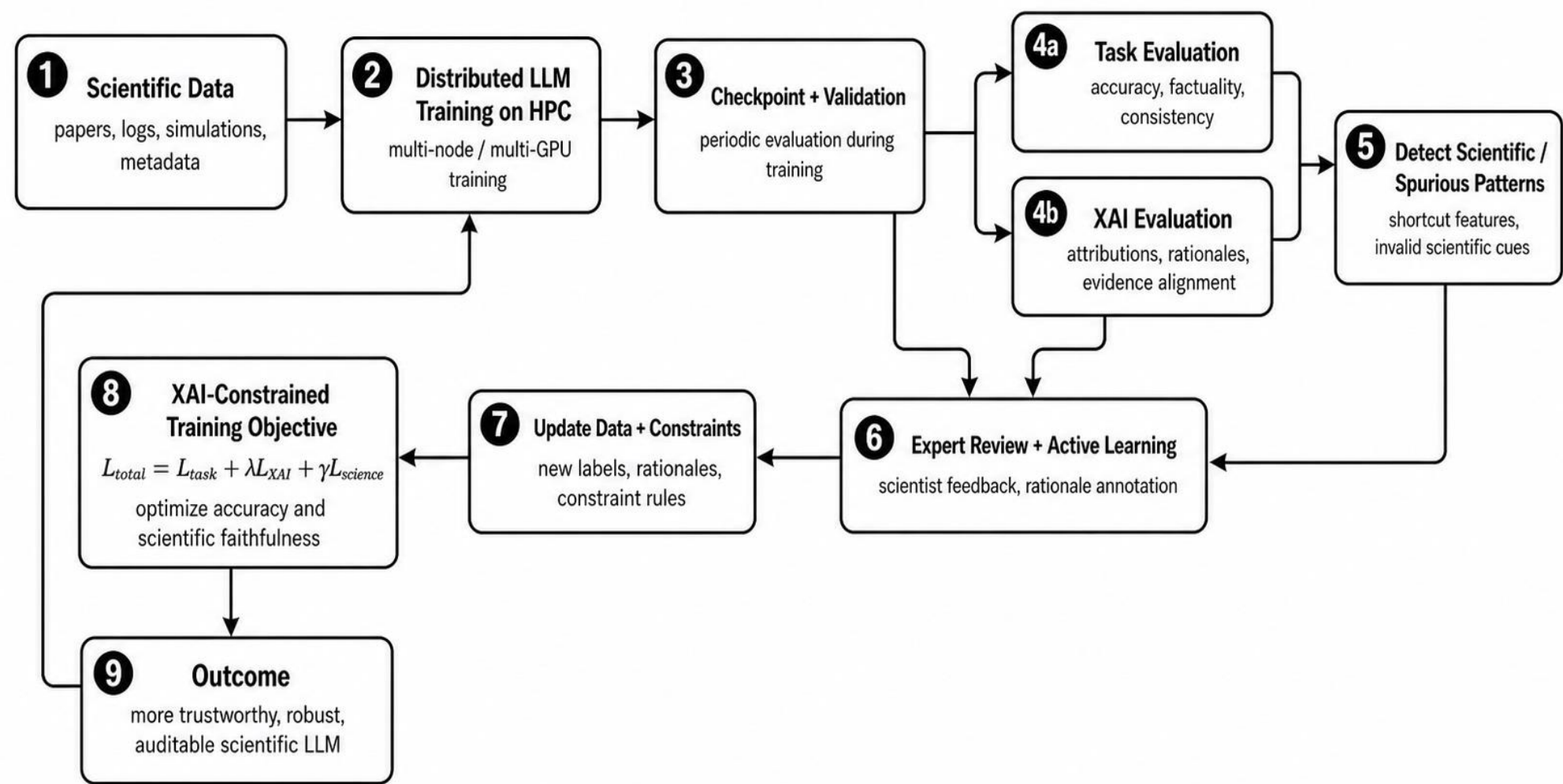
$\mathcal{L}_{science}$ penalizes scientific violations such as incorrect units, invalid equations, unsupported claims, or physically implausible reasoning.

λ and γ control the strength of explanation and scientific constraints.

Applying XAI as an Active Learning Constraint

- Scientific Data:** Collect scientific papers, logs, simulations, and metadata for training.
- Distributed LLM Training on HPC:** Train the LLM across multiple GPUs or compute nodes.
- Checkpoint + Validation:** Evaluate model behavior at scheduled training checkpoints.
- (a) Task Evaluation:** Measure accuracy, factuality, and consistency.
(b) XAI Evaluation: Assess attributions, rationales, and evidence alignment.
- Detect Scientific / Spurious Patterns:** Identify shortcut features or invalid scientific signals.
- Expert Review + Active Learning:** Use scientist feedback to select and correct examples.
- Update Data + Constraints:** Add new labels, rationales, and constraint rules.
- XAI-Constrained Training Objective:** Optimize accuracy while enforcing explanation and science constraints.
- Outcome:** Produce a more trustworthy, robust, and auditable scientific LLM.

XAI-Guided LLM Training in Scientific HPC Environments



Attribution Score (AS):

A model-agnostic probabilistic token attribution measure, using Bayes rule to invert the next-token log-probabilities so as to capture the model's internal representation of the distribution over token sequences. Independent of the details of the model's computational structure. Yields the conditional probability of the response given the prompt, and of the response given the prompt with a token marginalized away.

Sentence : "The water bottle didn't fit into the bag because it was too big." Question : "What is 'it' referring to?"
GEMMA-3-1B-IT Response: a) The water bottleb) The bagc) The water bottled)

Sentence : "The water bottle didn't fit into the bag because it was too big." Question : "What is 'it' referring to?"
GPT-NEO-1.3B Response: What is 'too big'?"The water bottle didn't fit into the bag because it

Sentence : "The water bottle didn't fit into the bag because it was too big." Question : "What is 'it' referring to?"
LLAMA-3-2-1B-INSTRUCT Response: The water bottle, the bag, or something else?"Answer: The water bottle. The sentence is

Sentence : "The water bottle didn't fit into the bag because it was too big." Question : "What is 'it' referring to?"
OLMO-2-0425-1B Response: (A) The water bottle. (B) The bag. (C) The water. <endofxtxt>

Sentence : "The water bottle didn't fit into the bag because it was too big." Question : "What is 'it' referring to?"
OLMO-2-0425-1B-INSTRUCT Response: ". Possible answers: 1. water bottle. 2. bag. 3. the water

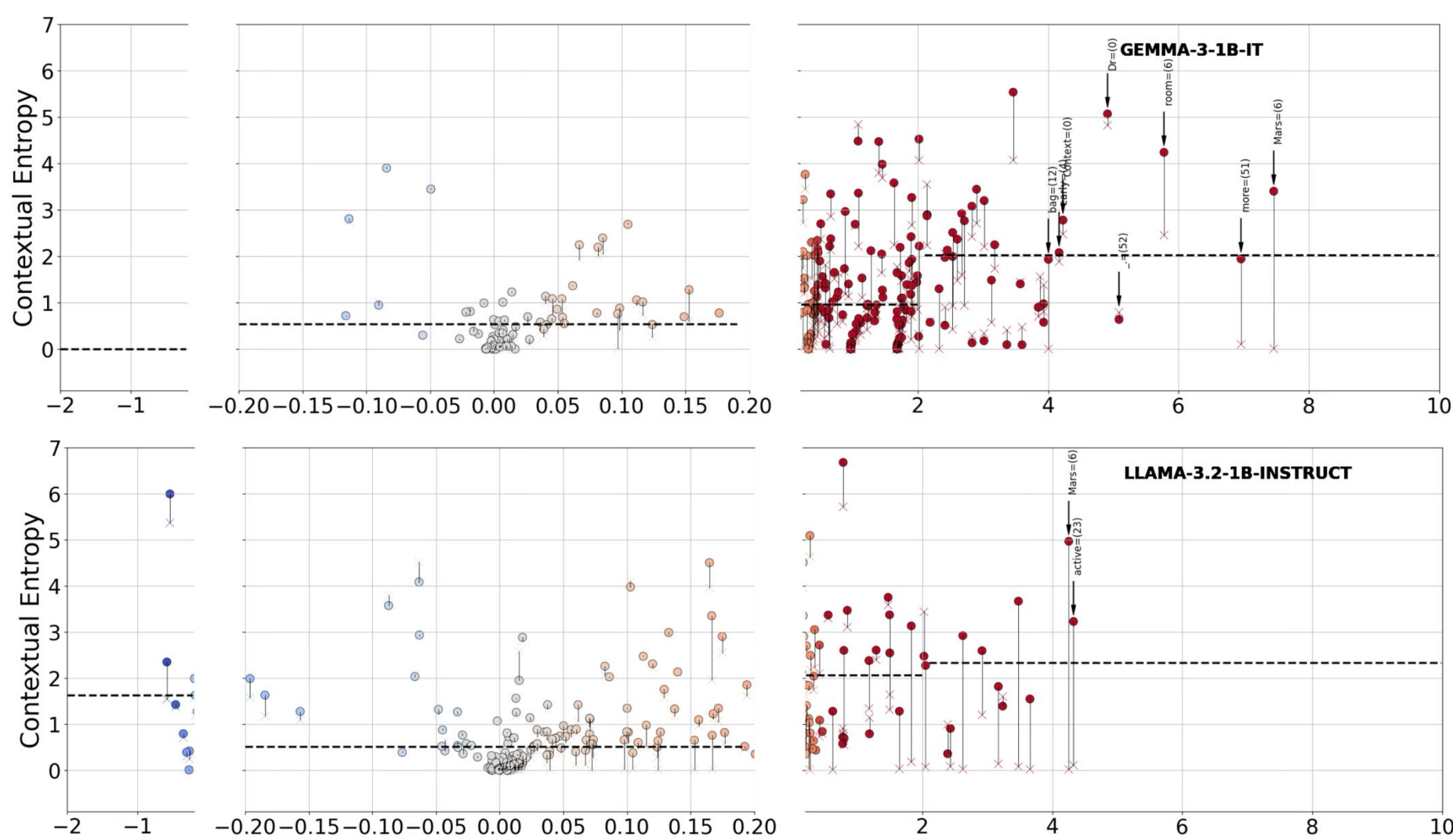
Pass age : "In 1997, NASA's Cassini space probe was launched to explore Saturn. It was active for twenty years, spending it's last 13 years orbiting Saturn." Question : "When was the Cassini launched?"
GEMMA-3-1B-IT Response: The answer is: 1997."<end_of_turn><end_of_turn><end_of_turn>

Pass age : "In 1997, NASA's Cassini space probe was launched to explore Saturn. It was active for twenty years, spending it's last 13 years orbiting Saturn." Question : "When was the Cassini launched?"
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Pass age : "In 1997, NASA's Cassini space probe was launched to explore Saturn. It was active for twenty years, spending it's last 13 years orbiting Saturn." Question : "When was the Cassini launched?"
LLAMA-3-2-1B-INSTRUCT Response: In 1997, NASA's Cassini space probe was launched to explore Saturn. It was active

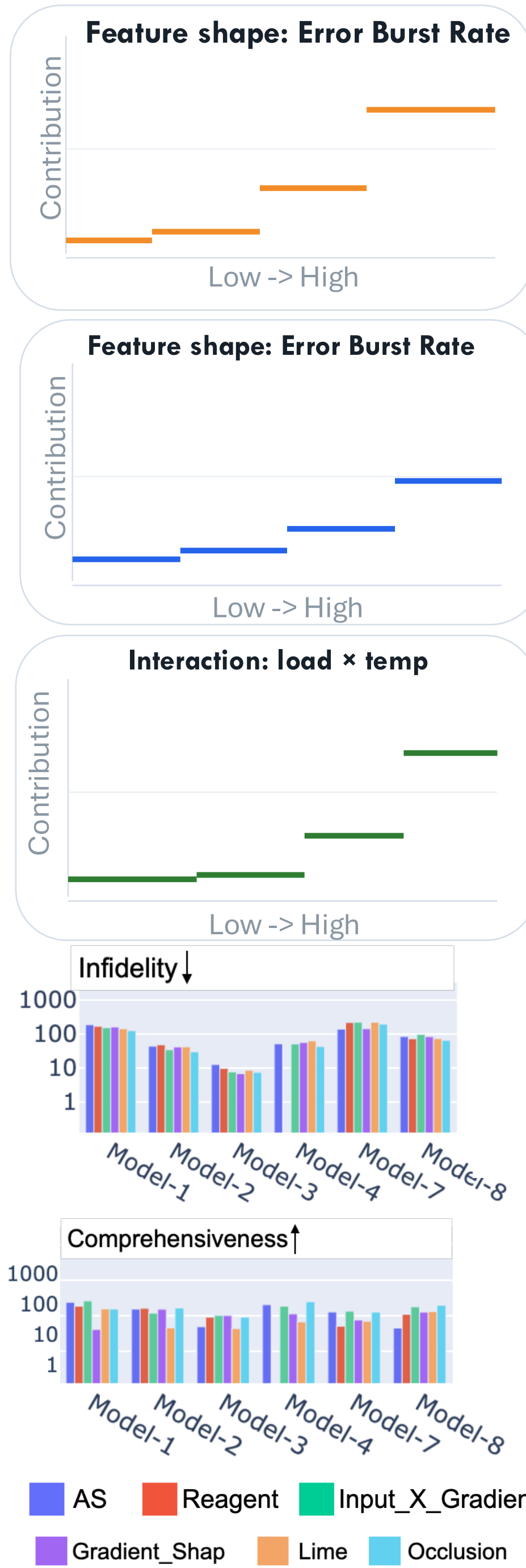
Pass age : "In 1997, NASA's Cassini space probe was launched to explore Saturn. It was active for twenty years, spending it's last 13 years orbiting Saturn." Question : "When was the Cassini launched?"
OLMO-2-0425-1B Response: "Is it the correct answer to the question "When was the Cassini space probe launched to explore

Pass age : "In 1997, NASA's Cassini space probe was launched to explore Saturn. It was active for twenty years, spending it's last 13 years orbiting Saturn." Question : "When was the Cassini launched?"
OLMO-2-0425-1B-INSTRUCT Response: " - A) 1997 - B) 2007 - C)



What the Explanation Exposes

If the model explains itself in ways that contradict scientific expectations, the feature layer or constraints are revised and the model is retrained.



References

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